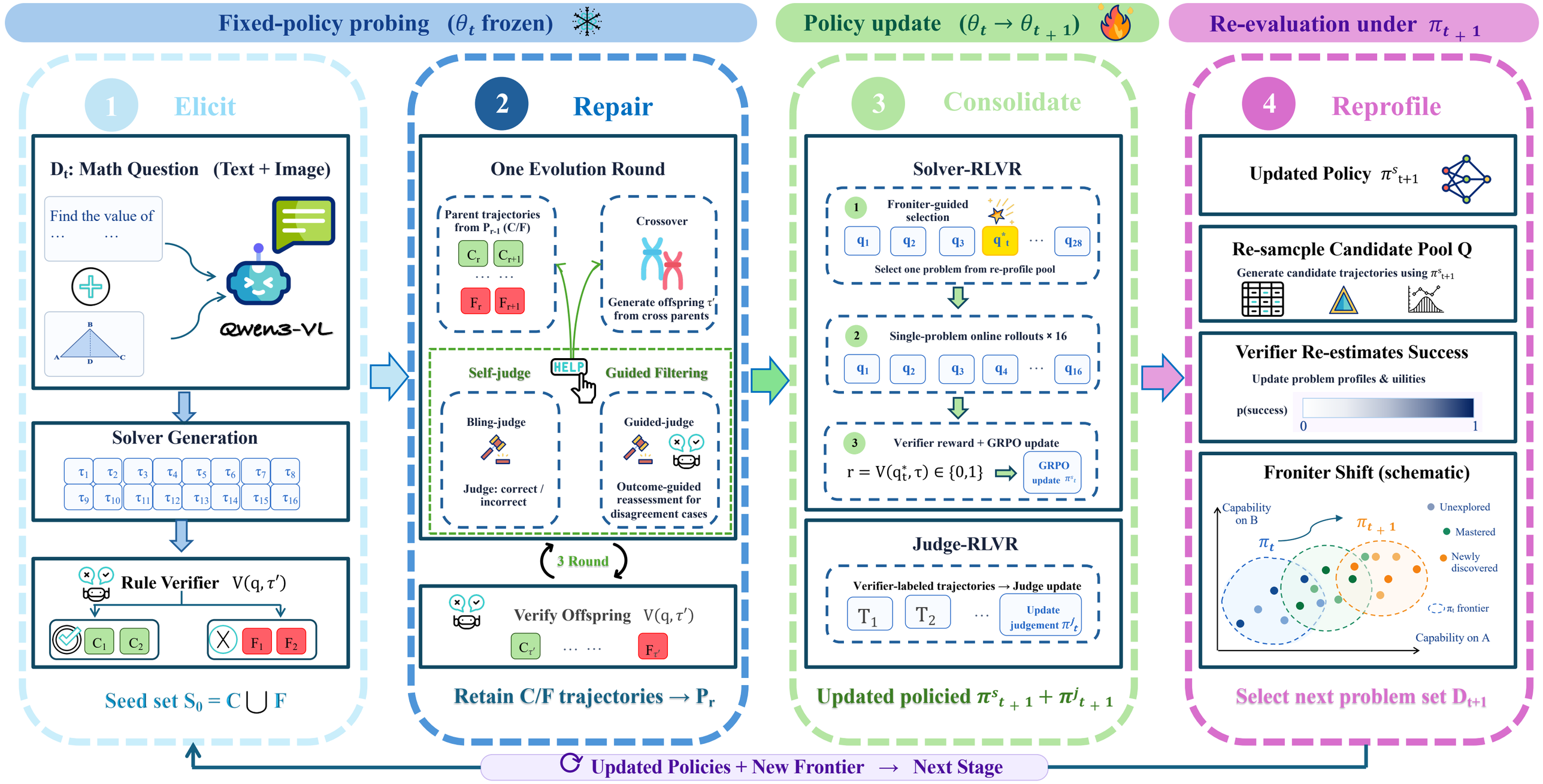

Fixed-policy probing (θ_t frozen)
Policy update ($\theta_t \rightarrow \theta_{t+1}$)
Re-evaluation under π_{t+1}
1 Elicit
D_t: Math Question (Text + Image)
Find the value of … …
Qwen3-VL
Solver Generation
τ_1 τ_2 τ_3 τ_4 τ_5 τ_6 τ_7 τ_8
τ_9 τ_{10} τ_{11} τ_{12} τ_{13} τ_{14} τ_{15} τ_{16}
Rule Verifier $V(q, \tau')$
C_1 C_2 F_1 F_2
Seed set $S_0 = C \cup F$
2 Repair
One Evolution Round
Parent trajectories from P_{r-1} (C/F)
C_r C_{r+1} … …
F_r F_{r+1}
Crossover
Generate offspring τ' from cross parents
Self-judge
HELP
Guided Filtering
Bling-judge
Judge: correct / incorrect
Guided-judge
Outcome-guided reassessment for disagreement cases
3 Round
Verify Offspring $V(q, \tau')$
$C_{\tau'}$ … … $F_{\tau'}$
Retain C/F trajectories → P_r
3 Consolidate
Solver-RLVR
1 Froniter-guided selection
q_1 q_2 q_3 q^*_t … q_{28}
Select one problem from re-profile pool
2 Single-problem online rollouts × 16
q_1 q_2 q_3 q_4 … q_{16}
3 Verifier reward + GRPO update
$r = V(q^*_t, \tau) \in \{0,1\}$
GRPO update π^s_t
Judge-RLVR
Verifier-labeled trajectories → Judge update
T_1 T_2 …
Update judgement π^j_t
Updated policied $\pi^s_{t+1} + \pi^j_{t+1}$
4 Reprofile
Updated Policy π^s_{t+1}
Re-samcple Candidate Pool Q
Generate candidate trajectories using π^s_{t+1}
Verifier Re-estimates Success
Update problem profiles & uilities
p(success)
0
1
Froniter Shift (schematic)
Capability on B
π_t
π_{t+1}
Unexplored
Mastered
Newly discovered
π_t frontier
Capability on A
Select next problem set D_{t+1}
Updated Policies + New Frontier → Next Stage